

第二次作业

$$W = u + iv$$

1. 设 $z = x + iy$ $W = u + iv$

$$u + iv = W = \frac{1}{x + iy} = \frac{x - iy}{x^2 + y^2}$$

$$\Rightarrow \begin{cases} u = \frac{x}{x^2 + y^2} \\ v = \frac{-y}{x^2 + y^2} \end{cases} \quad (x)$$

(3) $W = x, \quad u = \frac{x}{x^2 + y^2} = \frac{1}{2x}$
 $v = \frac{-y}{x^2 + y^2} = -\frac{y}{2x}$

$$u^2 + v^2 = 0 \quad \text{直线}$$

(4) $x^2 + y^2 = 4, \quad W = x, \quad u = \frac{x}{4}, \quad v = -\frac{y}{4}$
 $\Rightarrow u^2 + v^2 = \frac{x^2 + y^2}{16} = \frac{4}{16} = \frac{1}{4} \quad \text{为圆}$

(5) $(x-1)^2 + y^2 = 5 \Rightarrow x^2 + y^2 = 4 + 2x$

$$W = x \Rightarrow \begin{cases} u = \frac{x}{4 + 2x} \\ v = \frac{-y}{4 + 2x} \end{cases} \Rightarrow u^2 + v^2 = \frac{x^2 + y^2}{(4 + 2x)^2} = \frac{1}{4 + 2x}$$

$$\text{故 } 2u + 4(u^2 + v^2) = 2 \cdot \frac{x}{4 + 2x} + 4 \cdot \frac{1}{4 + 2x} = 1$$

$$\text{即 } (u + \frac{1}{4})^2 + v^2 = \frac{5}{16}, \quad \text{为圆}$$

2. $f(z) = \frac{1}{z} \left(\frac{z}{2} - \frac{\bar{z}}{2} \right)$
 $= \frac{1}{2i} \frac{z^2 - \bar{z}^2}{2} = \frac{1}{2i} \frac{(z - \bar{z})(z + \bar{z})}{2}$ 设 $z = u + iv$
 $= \frac{1}{2i} \cdot \frac{2iy \cdot 2x}{x^2 + y^2} = \frac{2xy}{x^2 + y^2}$

取 $y = kx \rightarrow 0$
 $\lim_{k \rightarrow 0} f(z) = \frac{2kx^2}{x^2 + k^2x^2} = \frac{2k}{1 + k^2}$ 与 k 有关
从而极限不存在

$$3. \frac{1}{2} g = kx \rightarrow 0$$

$$\lim_{kx \rightarrow 0} f(x) = \lim_{g=kx \rightarrow 0} \frac{kx^3}{1+k^2x^2} = \frac{k}{1+k^2} \text{ 与 } k \text{ 有关}$$

因此 $f(x)$ 在 $z=0$ 处不连续

$$5. 12, \quad u(x, y) = x+y, \quad v(x, y) = 0$$

$$\frac{\partial u}{\partial x} = 1 \neq \frac{\partial v}{\partial y} = 0 \quad \text{在 } z \text{ 平面上恒成立}$$

由 C-R 方程即证

$$13) \quad f(z) = \frac{1}{z} = \frac{1}{x-iy} = \frac{x+iy}{x^2+y^2}$$

$$u(x, y) = \frac{x}{x^2+y^2}, \quad v(x, y) = \frac{y}{x^2+y^2}$$

① $z=0$ 时, $f(z)$ 无意义

$$\textcircled{2} z \neq 0 \text{ 时, } \frac{\partial u}{\partial x} = \frac{\frac{x^2-y^2}{x^2+y^2}}{(x^2+y^2)^2}, \quad \frac{\partial u}{\partial y} = \frac{-2xy}{(x^2+y^2)^2}$$

$$\frac{\partial v}{\partial x} = \frac{-2xy}{(x^2+y^2)^2}, \quad \frac{\partial v}{\partial y} = \frac{\frac{x^2-y^2}{x^2+y^2}}{(x^2+y^2)^2}$$

由 C-R 方程

$$\begin{cases} \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \\ \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \end{cases} \Rightarrow \begin{cases} \frac{x^2-y^2}{x^2+y^2} = \frac{x^2-y^2}{x^2+y^2} \\ -2xy = +2xy \end{cases}$$

$$\Rightarrow \begin{cases} x^2 = y^2 \\ xy = 0 \end{cases} \Rightarrow x=y=0 \quad \text{不满足 } z \neq 0$$

从而 z 平面处处不可导

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$$b(2) \quad \begin{cases} \sqrt{x^2+y^2} (x+y) \\ x^2-y^2+2xy \end{cases} \quad |z| < 1$$

$$f(z) = \begin{cases} \lambda \sqrt{x^2+y^2}, & |z| < 1 \\ x^2-y^2, & |z| \geq 1 \end{cases}$$

$$\Rightarrow u(x,y) = \begin{cases} \frac{1}{2} \sqrt{x^2+y^2}, & |z| < 1 \\ x^2-y^2, & |z| \geq 1 \end{cases}$$

$$v(x,y) = \begin{cases} \frac{1}{2} \sqrt{x^2+y^2}, & |z| \geq 1 \\ 2xy, & |z| < 1 \end{cases}$$

$$① \quad |z| < 1 \text{ 时, } \frac{\partial u}{\partial x} = \frac{x}{2\sqrt{x^2+y^2}}, \quad \frac{\partial v}{\partial y} = \frac{xy}{\sqrt{x^2+y^2}}$$

$$\frac{\partial v}{\partial x} = \frac{y^2}{\sqrt{x^2+y^2}}, \quad \frac{\partial v}{\partial y} = \frac{x^2+y^2}{\sqrt{x^2+y^2}}$$

$$f(z) = x^2 + y^2 \quad \begin{cases} \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \\ \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y} \end{cases} \Rightarrow \begin{cases} 2x^2+y^2 = x^2+2y^2 \\ xy = -xy \end{cases}$$

$\Rightarrow \begin{cases} x^2=y^2 \\ xy=0 \end{cases} \Rightarrow x=y=0$ 这说明只有 $z=0$ 满足柯西条件, 而 $z=0$ 的任意邻域均有点

不解析, 故 $z=0$ 不解析

$$② \quad |z| > 1 \text{ 时, } \frac{\partial u}{\partial x} = 2x = \frac{\partial v}{\partial y}$$

解析

③ $|z|=1$ 时, 其邻域内均包含 $|z|<1$ 的点, 故不解析
任意

综上, 解析区域为 $|z|>1$

$$7.12) u(x, y) = e^x (x \cos y - y \sin y)$$

$$v(x, y) = e^x (y \cos y + x \sin y)$$

$$\frac{\partial u}{\partial x} = e^x (x \cos y - y \sin y + \cos y)$$

$$\frac{\partial u}{\partial y} = e^x (-x \sin y - \sin y - y \cos y)$$

$$\frac{\partial v}{\partial x} = e^x (\sin y + y \cos y + x \sin y)$$

$$\frac{\partial v}{\partial y} = e^x (-y \sin y + \cos y + x \cos y)$$

$$\Rightarrow \begin{cases} \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \\ \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \end{cases} \text{ 恒成立, 故全平面解析}$$

$$f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}$$

$$= e^x (x \cos y - y \sin y + \cos y) + i e^x (\sin y + y \cos y + x \sin y)$$

$$8.13) u(x, y) = C, \text{ 为常数}$$

$$\text{则 } \frac{\partial u}{\partial x} = 0, \frac{\partial u}{\partial y} = 0$$

$$\text{由 C-R 方程 } \frac{\partial v}{\partial y} = \frac{\partial u}{\partial x} = 0, \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y} = 0$$

$$\text{从而 } v(x, y) = C_2 \text{ 也为常数}$$

$$f(z) = u + iv = C_1 + iC_2 \text{ 为常数}$$

$$(b) \text{ ①若 } \arg f(z) = \frac{\pi}{2}, \text{ 则 } \operatorname{Re} f(z) = 0 \text{ 由 (b) 知为常数}$$

$$\text{②若 } \arg f(z) \neq \frac{\pi}{2}, \text{ 可设 } v(x, y) = k u(x, y)$$

$$\text{则 } \frac{\partial v}{\partial x} = k \frac{\partial u}{\partial x}, \frac{\partial v}{\partial y} = k \frac{\partial u}{\partial y}$$

$$\text{代入 C-R 方程}$$



$$\begin{cases} \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \\ \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \end{cases} \Rightarrow \begin{cases} \frac{\partial u}{\partial x} = k \frac{\partial u}{\partial y} \\ \frac{\partial u}{\partial y} = -k \frac{\partial u}{\partial x} \end{cases}$$

$$\Rightarrow (k^2 + 1) \frac{\partial u}{\partial x} = -k^2 \frac{\partial u}{\partial x}, \text{ 即 } (k^2 + 1) \frac{\partial u}{\partial x} = 0$$

$$\Rightarrow \frac{\partial u}{\partial x} = 0, \text{ 进而 } \frac{\partial u}{\partial y}, \frac{\partial v}{\partial x}, \frac{\partial v}{\partial y} \text{ 均为 } 0$$

因此, $u(x, y), v(x, y)$ 均为常数

故 $f(z)$ 为常数

10.

$$(1) f(z) = \frac{1}{z^2 - 3z + 2} = \frac{1}{(z-1)(z-2)} = -\frac{1}{z-1} + \frac{1}{z-2}$$

解析区域为 $\{z \mid z \neq 1 \text{ 且 } z \neq 2\}$

$$f'(z) = -\left(\frac{1}{z-1}\right)' + \left(\frac{1}{z-2}\right)' = \frac{1}{(z-1)^2} - \frac{1}{(z-2)^2}$$

$$(2) g(z) = \frac{1}{z^3 + a}$$

$$z^3 + a \neq 0 \Rightarrow z \neq \sqrt[3]{a} \left(\cos \frac{\pi + 2k\pi}{3} + j \sin \frac{\pi + 2k\pi}{3} \right),$$

$k = 0, 1, 2$, 此即解析区域

$$g'(z) = \frac{-3z^2}{(z^3 + a)^2}$$

常数